Applications of similar triangles

1 $u = 36$

2

a

i 2

ii $x = 14$

b

i 3

ii $g = 8$

3 $h = 36$ m

4 $h = 96$ cm

5 $h = 14.7$ m

6 $h = 3.1$ m

7 $h = 1.4$ m

8 $d = 17.1$

9 $h = 11.1$

10 $h = 4.8$ m

11 $w = 77$ m

12 $L = 4.7$

13

a $x = \frac{7}{2}$

b $r = 9.3$

14 18 ft

15 225 in

16 $h = 15$

17 $L = 5.3$

18

a 180 m

b 120 m

Additional questions

19

a $x = 5$

b $x = 5$

c $y = \frac{4}{3}$

d $y = \frac{1}{2}$

e $x = 14$
$y = 22.4$

f $x = 13,\ y = \frac{60}{13}$

g $x = \frac{5\sqrt{39}}{8},\ y = \sqrt{39}$

20 The Super Slide is 143.4375 ft tall.

21 The flag pole is about 17.3 ft tall.

22

a $\overline{CD} = 3\sqrt{2}$

b $\overline{CB} = 3\sqrt{3}$

c $\overline{AC} = 3\sqrt{6}$

23 The height of the roof is about 4.33 m.

24 Lot 031 will have the lowest list price since it has the least amount of beachfront at 52.68 yd. Lot 033 will have the highest list price since it has the most beachfront at 66.95 yd.

25

a Yes. Since a pair of legs on the right triangles is in the same ratio as the two hypotenuses, the triangles are similar using the Hypotenuse-Leg test for similarity.

b Yes. Since a pair of legs on the right triangles is in the same ratio as the two hypotenuses, the triangles are similar using the Hypotenuse-Leg test for similarity.

c Yes. Since a pair of legs on the right triangles is in the same ratio as the two hypotenuses, the triangles are similar using the Hypotenuse-Leg test for similarity.

27 $TU = 3$

$TV = \frac{45}{17}$

$RS = 1.26$

$RX = \frac{1719}{170}$

$PX = 15$

$PQ = \frac{3069}{170}$

28 The hypotenuse has a length of $6\sqrt{5}$ in.